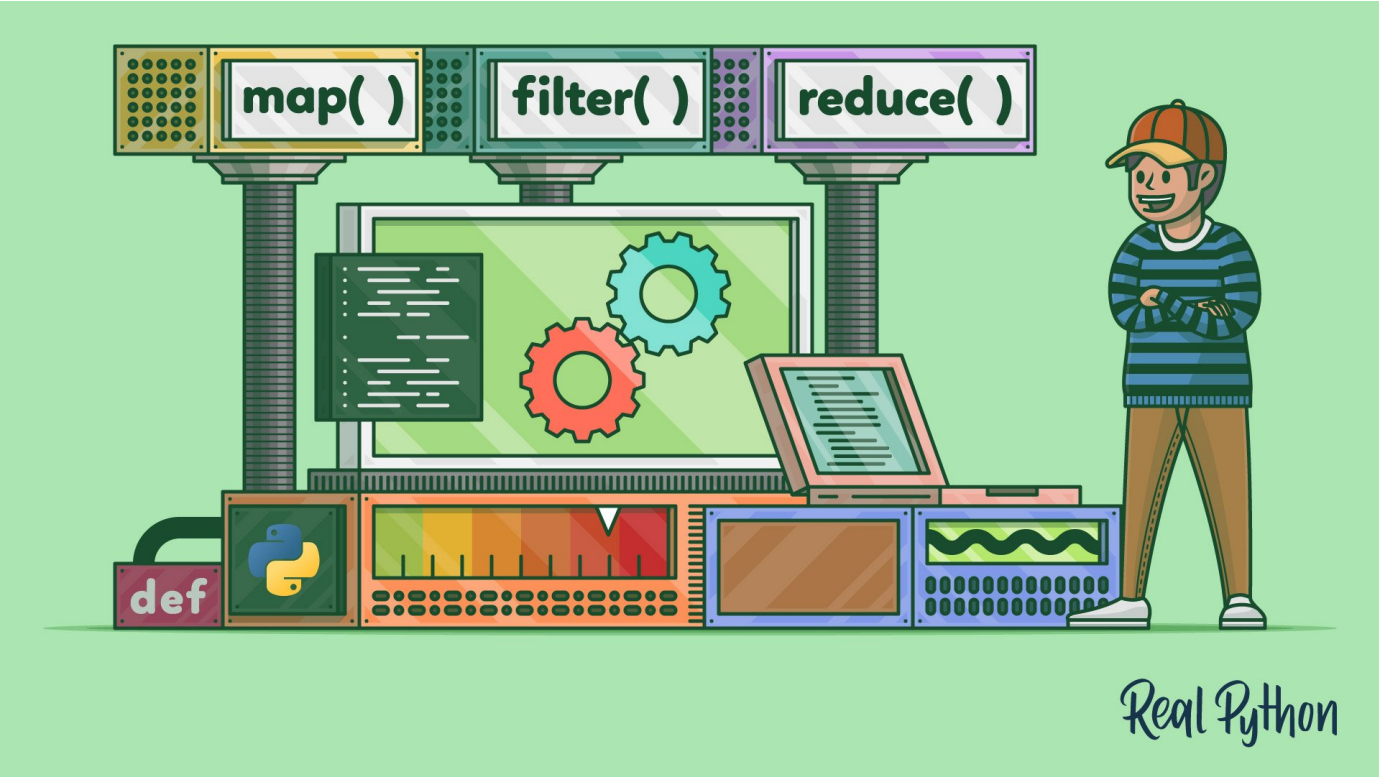
USING FUNCTIONAL PROGRAMMING IN PYTHON



YOU WILL LEARN ABOUT:

1. Functional programming
2. Functions as objects
3. `map()`
4. `filter()`
5. `reduce()`

VERSIONS

Note:

- Sample code tested using:
  - Python 3.14

OVERVIEW

- Functional programming is an approach that composing functions together
- Python doesn’t require a functional programming approach but does support it
- Functions are objects
  - Assign values to them
  - Pass them as references
  - Return references to them
- Python comes with a selection of functional programming tools

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Functional programming approach

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PROGRAMMING APPROACHES

```
# Procedural
x =
y =
prin
```

```
# Object-Oriented
class
    d
    d
```

```
# Functional
def work(values, callback_fn):
    total = sum(values)
    callback_fn(total)
```

FUNCTIONAL PROGRAMMING

- Functional programming typically needs:
  - Functions being able to be passed as arguments (not just executed)
  - Functions being able to return a function reference
- In Python functions are first-class citizens
  - Everything in Python is an object, even functions
  - Share basic properties of strings, integers, etc

WHY FUNCTIONAL PROGRAMMING?

- Pure functional programming avoids side effects
- Compose functions together
- As state tends to be passed (no side effects) it is self contained
  - Transparent behavior
  - Parallelizable

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Functions as objects

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Lambdas

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LAMBDAS

- Functions have overhead in Python
- Sometimes you want something lighter weight
- Lambdas are:
  - Anonymous
  - Can be used inline
  - Restricted to a single statement

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`map()`

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PYTHON'S `map()`

```
map(callable_ref, iterable)
```

- Apply a function to each item in an iterable

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`filter()`

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PYTHON'S `filter()`

```
filter(callable_ref, iterable)
```

- Filter out results from an iterable

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`reduce()`

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PYTHON'S `reduce()`

```
reduce(callable_ref, iterable)
⋮
output = iterable[0]
for item in iterable[1:]:
    output = callable_ref(output, item)
```

- Compose results based on previous result and the next item in the iterable

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Summary

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SUMMARY

- Functional programming chains function calls together
- Requires the use of functions as both data and invocation targets
- Functions in Python are objects
- Python lambdas are anonymous, single statement functions
- Three common functional programming utilities are:
  - `map()`
  - `filter()`
  - `reduce()`

MORE AT REAL PYTHON

- Using Python Lambda Functions
  - https://realpython.com/courses/python-lambda-functions/
  - https://realpython.com/python-lambda/ (tutorial)
- `map()`, `filter()`, and `reduce()`
  - https://realpython.com/python-map-function/ (tutorial)
  - https://realpython.com/python-filter-function/ (tutorial)
  - https://realpython.com/python-reduce-function/ (tutorial)
- Speed Up Python With Concurrency
  - https://realpython.com/courses/speed-python-concurrency/
  - https://realpython.com/python-concurrency/ (tutorial)

Thanks!